

FOSD — Unit 7

File System

GATE CSE PYQ source-page collection • 2017–2026

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- The question pages below are taken directly from the original source PDFs supplied in this conversation.
- Original question wording, options, tables, code, and figures are preserved on those pages.
- A complete original page is retained whenever a selected question depends on a figure/diagram, so the required visual context is not lost.
- The selected question numbers are listed below; neighboring questions may remain visible because the original page is preserved.
- The supplied 2017 Set-1 PDF is corrupted/unreadable in the uploaded ZIP, so no 2017 question is fabricated or substituted here.

Included source pages

CS1-2021 — source page 18 — selected Q15

CS2022 — source page 40 — selected Q63

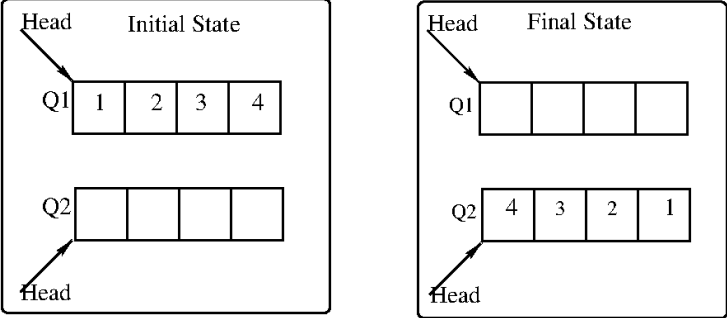
CS12025 — source page 51 — selected Q51

CS2-2026 — source page 38 — selected Q53

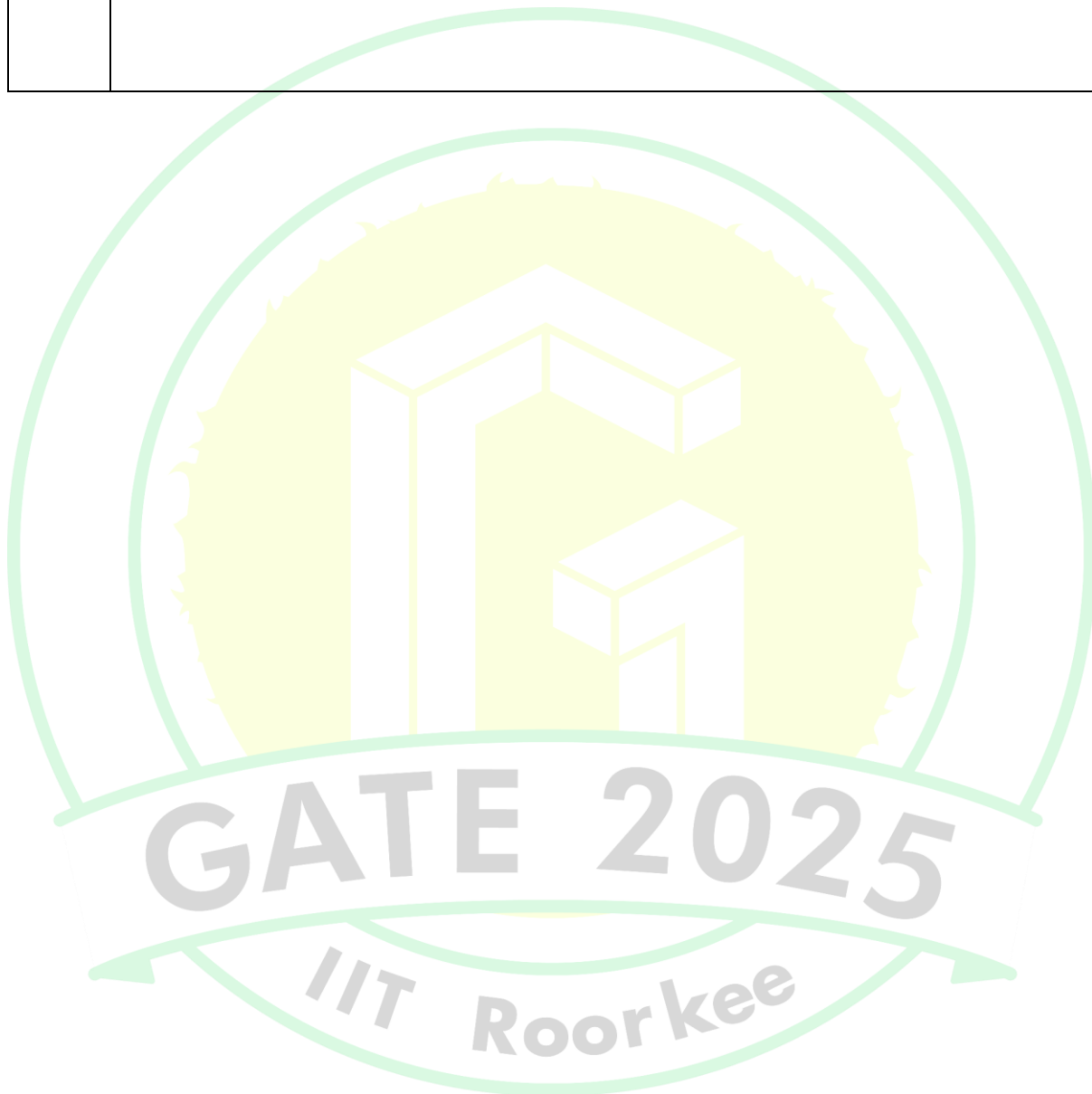


Q.15	Consider a linear list based directory implementation in a file system. Each directory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory <code>foo</code>. Which of the following operations will necessarily require a full scan of <code>foo</code> for successful completion?
(A)	Creation of a new file in <code>foo</code>
(B)	Deletion of an existing file from <code>foo</code>
(C)	Renaming of an existing file in <code>foo</code>
(D)	Opening of an existing file in <code>foo</code>

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Q.62	Consider the queues Q_1 containing four elements and Q_2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue(Q, element) and Dequeue(Q). The minimum number of Enqueue operations on Q_1 required to place the elements of Q_1 in Q_2 in reverse order (shown as the Final State in the figure) without using any additional storage is _____.
	
Q.63	Consider two files systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B, respectively, then the value of $n_A + n_B$ is _____.
Q.64	Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string 7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is _____.

Q.51	A disk of size 512M bytes is divided into blocks of 64K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a <i>NULL</i> pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, $1K = 2^{10}$ and $1M = 2^{20}$. The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer)



Q.52	Consider a system with a processor and a 4 KB direct mapped cache with block size of 16 bytes. The system has a 16 MB physical memory. Four words P, Q, R, and S are accessed by the processor in the same order 10 times. That is, there are a total of 40 memory references in the sequence P, Q, R, S, P, Q, R, S,... Assume that the cache memory is initially empty. The physical addresses of the words are given below (1 word = 1 byte). P: 0x845B32, Q: 0x845B26, R: 0x845B36, S: 0x846B32 Which of the following statements is/are true? <i>Note: $1K=2^{10}$ and $1M=2^{20}$</i>
(A)	Every access to P results in a cache miss
(B)	Every access to R results in a cache hit
(C)	Every access to Q results in a cache miss
(D)	Except the first access to S, all subsequent accesses to S result in cache hits
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Q.53	To keep track of free blocks in a file system, one of the two approaches is generally used – using bitmaps (bit vectors) or using linked lists. Consider that the linked list approach is used to keep track of free blocks in a file system. Assume that the disk size is 16 GB, block size is 2 KB, and block numbers used are 32-bit long. A single pointer of size 4 bytes is used in each block of the list to point to the next block of the list. The number of blocks required to hold the free disk block numbers is _____. (<i>answer in integer</i>) <i>Note: $1K=2^{10}$ and $1G=2^{30}$</i>